3

Environmental Impact Assessment and Sustainable Development

3.1 INTRODUCTION

The present era of fast development and growth is aimed at raising the quality of human life by providing greater opportunities for employment, better provisions of basic amenities and comforts, healthy environment ensuring physical and mental well-being of humans. Also growth and development lead to several environmental problems like pollution of the air, water and soil, depletion of natural resources, energy crisis, occupational health problems, and global problems like climate change, ozone layer depletion, and loss of biodiversity. Thus, development is bound to have certain environmental impacts. It was about 40 years back when it was realized that before a development project is started, prediction and assessment of its impacts should be done, so that measures could be taken to minimize those impacts. This concept was formulated as a methodical procedure known as **Environmental Impact Assessment (EIA).**

Further, for achieving the goals of real improvement in the quality of human life, development should be based on sustainability principles. Thus, **sustainable development** aims at growth with judicious use of resources and causing minimum damage to the environment.

3.2 ENVIRONMENTAL IMPACT ASSESSMENT (EIA)

Environmental Impact Assessment (EIA) is a procedure to plan some developmental activity with well-defined environmental goals so that damage due to the activity both during developmental stage and production stage have minimum impact on the natural system and the population in the area.

The National Environmental Policy Act (NEPA) U.S.A. in 1969 first of all provided the guidelines for environmental impact assessment through Council for Environmental Quality (CEQ).

In India, the gazette notification on EIA was issued in 1994 vide which the Ministry of Environment and Forests provided guidelines for project proponents to have EIA and prepare an Environmental Impact Statement prior to clearance of the project.

3.2.1 Goals of EIA

(*i*) To fulfill the responsibilities towards the coming generations as trustees of environment.
(*ii*) To assure safe, healthy, productive, aesthetically as well as culturally pleasing surroundings.
(*iii*) To provide widest range of beneficial uses of environment without degradation or risk to health.
(*iv*) To preserve historical, cultural and natural heritage.
(*v*) To achieve a balance between population and resource use for a good standard of living.
(*vi*) To ensure sustainable development with minimal environmental degradation.

3.2.2 Environmental Impact Statement (EIS)

The EIS is prepared by the project proponents at the time of submission of the proposal, which is known as the *draft EIS.* After evaluation and review by the Impact Assessment Agency, the *final EIS* is prepared.

The following points are usually incorporated while preparing the EIS:

- Effect on land including land degradation and subsistence.
- Deforestation and compensatory afforestation.
- Air pollution and dispersion along with possible health effects.
- Water pollution including surface water and ground water pollution.
- Noise pollution due to the project.
- Loss of flora and fauna due to the project during construction.
- Socio-economic impacts including displacement of native people, cultural loss and health aspects.
- Risk analysis and disaster management plan.
- Recycling and reduction of waste.
- Efficient use of inputs including energy and matter.

EIA is done with an aim to select the best alternative through which adverse impact on the environment can be nullified or minimized without compromising with the economic and social benefits of the developmental project.

Four types of alternatives are considered:

(*i*) **Alternative technologies** providing options with maximum energy efficiency and minimal wastage.
(*ii*) **Alternative mitigating or controlling mechanisms** through which recycling of by-products or reduction of emissions can take place.
(*iii*) **Alternate phasing** to work out if phasing of the project is possible instead of one stroke development to avoid drastic impact.
(*iv*) **Alternate site** for the proposed project.

However, the most important alternative taken into consideration in EIA is the impact assessment at alternative sites *i.e.,* which of the site I or II or III located in different natural area would have the least impact of the development project, and that site is selected for the development project.

Thus, the main purpose of EIA is precisely to estimate the type and level of damage caused to natural environment in a well-defined time scale so that remedial measures can be initiated on those aspects requiring action at the right time.

3.2.3 EIA Methodology

The basic steps followed in EIA are screening, scoping, base line data, impact identification, prediction, evaluation, mitigation, EIS preparation, review and environment audit, involving public participation at various stages, as shown in Fig. 3.1.

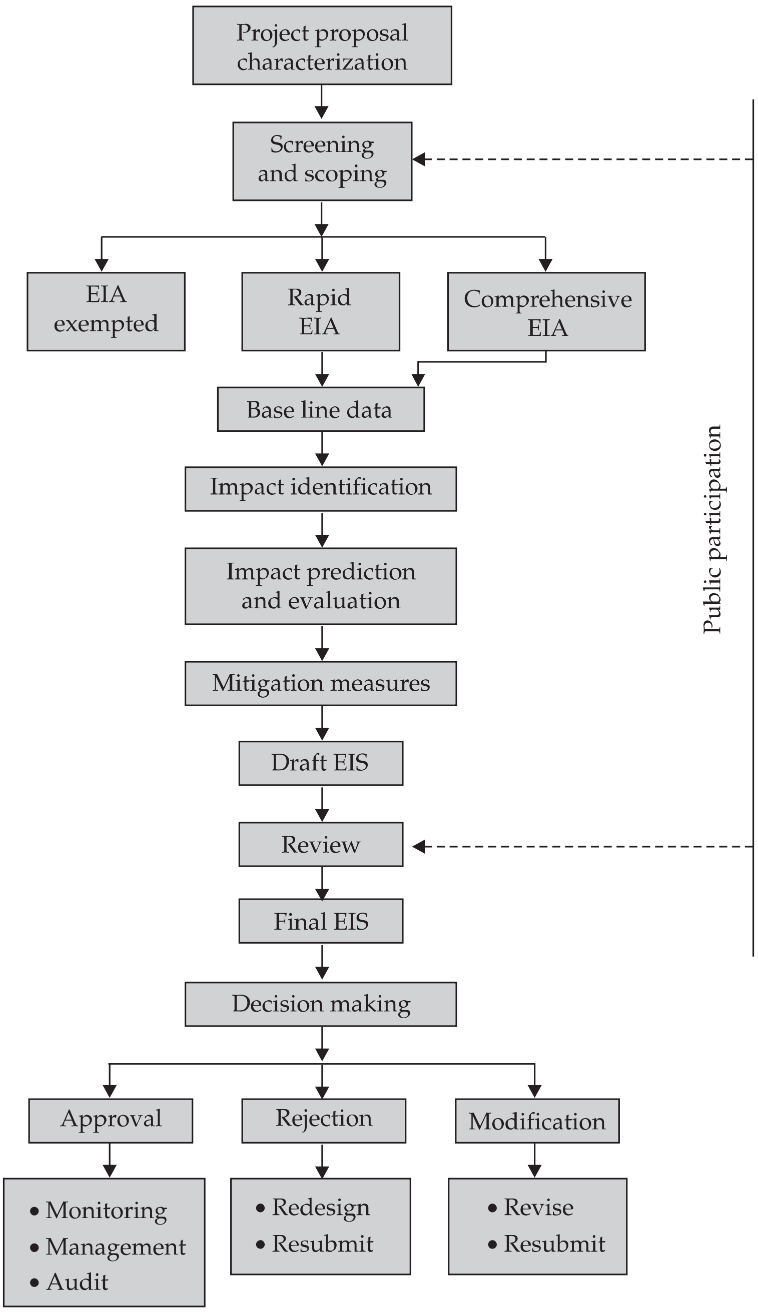

Fig. 3.1. EIA methodology flow-chart.

(*i*) **Screening** is done to see whether the project needs an EIA for clearance or not. Further, there are some prohibited areas where generally development projects are not allowed *e.g.,* Coastal Regulation Zone (CRZ), Dahanu Taluka in Maharashtra, Aravalli range, Reserve forests etc.

(*ii*) **Scoping** involves determination of the extent of EIA required for the project. Depending upon the project, basically two types of EIA may be carried out. When the EIA report is based on a single season data (other than monsoon period), it is called *rapid EIA*. When the EIA report is based on detailed seasonal data, it is called *comprehensive EIA*.

(*iii*) **Baseline data** gives a holistic picture of the overall environmental setting of the project location showing any significant environmental items prior to initiation of the action; any potentially critical environmental changes and information about the site to the decision makers and reviewers, who might be unfamiliar with the general location of the project area.

The following environmental parameters are usually considered while preparing the baseline data:

(*a*) Site location and topography.

(*b*) Regional demography – population distribution within 10 and 50 kilometer radius; land-use and water-use pattern.

(*c*) Regional landmarks like historical and cultural heritage in the area. For this archaeological or state register can be checked.

(*d*) Geology – Groundwater and surface water resources are quantified; water, quality, pollution sources etc. are studied.

(*e*) Hydrology – Groundwater and surface water resources are quantified; water, quality, pollution sources etc. are studied.

(*f*) Meteorology – Temperature extremes, wind speed and direction, dew point, atmospheric stability, rainfall, storms etc. are recorded.

(*g*) Ecology – The flora, fauna, endangered species, successional stage etc. are enlisted.

For a particular project, some of the parameters may be important while for others, some other parameters could be important.

(*iv*) **Impact identification:** It includes the details of project characters and baseline environmental characteristics to ensure the identification of full range of environmental impacts.

During identification process, the positive and negative, direct and indirect significant and insignificant impacts are considered.

(*v*) **Impact prediction:** Here magnitude of changes going to occur due to the project are predicted by using mathematical models or mass balance models.

(*vi*) **Impact evaluation:** Impact evaluation is done by considering the costs and benefits of the project. Long-term effects and side-effects of the project are also evaluated. Indirect valuation of environmental parameters are also done. *e.g.* loss of a rare species, degradation of a lake etc.

(*vii*) **Mitigation:** Once the impacts are predicted and evaluated, mitigation measures are to be suggested to avoid, reduce or rectify the adverse changes due to the project.

Review and a draft impact statement is prepared at this stage.

(*viii*) **Decision analysis:** Public participation is involved by arranging group discussion or by adopting questionnaire method to arrive at a decision about the project and its evaluation.

(*ix*) **Environmental impact statement (EIS):** Based on the data obtained and review suggestions a final EIS is prepared as per the format provided by the Ministry of Environment and Forests in our country.

The EIS clearly mentions the objectives of the project, its environmental impacts, impacts that are unavoidable, mitigation measures to minimize the impacts, alternatives to the proposed action etc.

(*x*) **Environmental audit:** It compares the impacts predicted in EIS before the project was started and actual impacts after implementation of the project.

3.3 SUSTAINABLE DEVELOPMENT

Human beings live in both natural and social world. Our technological development has strong impacts on the natural as well as the social components. When we talk of development, it cannot be perceived as development only for a privileged few who would have a high standard of living and would derive all the benefits. Development also does not mean an increase in the GNP (Gross National Product) of a few affluent nations. Development has to be visualized in a holistic manner, where it brings benefits to all, not only for the present generation, but also for the future generations.

There is an urgent need to inter-link the social aspects with development and environment. In this unit we shall discuss various social issues in relation to environment.

Sustainable development is defined as "meeting the needs of the present without compromising the ability of future generations to meet their own needs." This definition was given in Brundtland Commission Report, ''Our Common Future'', by the Norwegian Prime Minister, G.H. Brundtland, who was also the Director of World Health Organisation (WHO). Today sustainable development has become a buzz word and hundreds of programmes have been initiated in the name of sustainable development. If you want to test whether or not a proposal will achieve the goals of sustainability just try to find out the following:

– Does it protect our biodiversity?

– Does it prevent soil erosion?

– Does it slow down population growth?

– Does it increase forest cover?

– Does it cut off the emissions of CFC, SO_x, NO_x and CO_2?

– Does it reduce waste generation and does it bring benefits to all?

These are only a few parameters for achieving sustainable growth.

Until now development has been human-oriented, that too mainly, for a few rich nations. They have touched the greatest heights of scientific and technological development, but at what cost? The air we breathe, the water we drink and the food we eat have all been badly polluted. Our natural resources are just dwindling due to over exploitation. If growth continues in the same way, very soon we will be facing a "doom's day"—as suggested by Meadows and co-workers in their world famous academic report "***The Limits to Growth***". This is unsustainable development which will lead to a collapse of the inter-related systems of this earth.

Although the fears about such unsustainable growth and development started in 1970's, yet a clear discussion on sustainable development emerged on an international level in 1992, in the **UN Conference on Environment and Development (UNCED)**, popularly known as The **Earth Summit**, held at Rio de Janeiro, Brazil. The Rio Declaration aims at "*a new and equitable global partnership through the creation of new levels of cooperation among states* ...." Out of its five significant agreements **Agenda-21** proposes a global programme of action on sustainable development in social, economic and political context for the 21st century.

This was followed by **UN World Summit on Sustainable Develpment (WSSD)** in Johannesburg, South Africa in 2002 which emphasized on national strategies for sustainable development.

The key aspects for sustainable development are:

(*a*) **Inter-generational equity:** This emphasizes that we should minimize any adverse impacts on resources and environment for future generations *i.e.* we should hand over a safe, healthy and resourceful environment to our future generations. This can be possible only if we stop over-exploitation of resources, reduce waste discharge and emissions and maintain ecological balance.

(*b*) **Intra-generational equity:** This emphasizes that the development processes should seek to minimize the wealth gaps within and between nations. The Human Development Report of United Nations (2001) emphasizes that the benefits of technology should seek to achieve the goals of intra-generational equity. The technology should address the problems of the developing countries, producing drought tolerant varieties for uncertain climates, vaccines for infectious diseases, clean fuels for domestic and industrial use. This type of technological development will support the economic growth of the poor countries and help in narrowing the wealth gap and lead to sustainability.

Measures for Sustainable Development: Some of the important measures for sustainable development are as follows:

(*i*) **Using appropriate technology** is one which is locally adaptable, eco-friendly, resource-efficient and culturally suitable. It mostly involves local resources and local labour. Indigenous technologies are more useful, cost-effective and sustainable. Nature is often taken as a model, using the natural conditions of that region as its components. This concept is known as "*design with nature*".

The technology should use less of resources and should produce minimum waste.

(*ii*) **Reduce, Reuse, Recycle approach:** The **3-R approach** advocating minimization of resource use, using them again and again instead of passing it on to the waste stream and recycling the materials goes a long way in achieving the goals of sustainability. It reduces pressure on our resources as well as reduces waste generation and pollution.

(*iii*) **Promoting environmental education and awareness:** Making environmental education the centre of all learning process will greatly help in changing the thinking pattern and attitude of people towards our earth and the environment. Introducing subject right from the school stage will inculcate a feeling of belongingness to earth in small children. 'Earth thinking' will gradually get incorporated in our thinking and action which will greatly help in transforming our lifestyles to sustainable ones.

(*iv*) **Resource utilization as per carrying capacity:** Any system can sustain a limited number of organisms on a long-term basis which is known as its **carrying capacity**. In case of human beings, the carrying capacity concept becomes all the more complex. It is because unlike other animals, human beings, not only need food to live, but need so many other things to maintain the quality of life.

Sustainability of a system depends largely upon the carrying capacity of the system. If the carrying capacity of a system is crossed (say, by over exploitation of a resource), environmental degradation starts and continues till it reaches a point of no return.

Carrying capacity has two basic components:

- **Supporting capacity** *i.e.* the capacity to regenerate
- **Assimilative capacity** *i.e.* the capacity to tolerate different stresses.

In order to attain sustainability it is very important to utilize the resources based upon the above two properties of the system. Consumption should not exceed regeneration and changes should not be allowed to occur beyond the tolerance capacity of the system.

(*v*) **Improving quality of life including social, cultural and economic dimensions:** Development should not focus just on one section of already affluent people. Rather it should include sharing of benefits between the rich and the poor. The tribal, ethnic people and their cultural heritage should also be conserved. Strong community participation should be there in policy and practice. Population growth should be stabilized.

Thus sustainable development can occur by integrating social, scientific and ecological dimensions at regional and global level, as illustrated in Fig. 3.2.

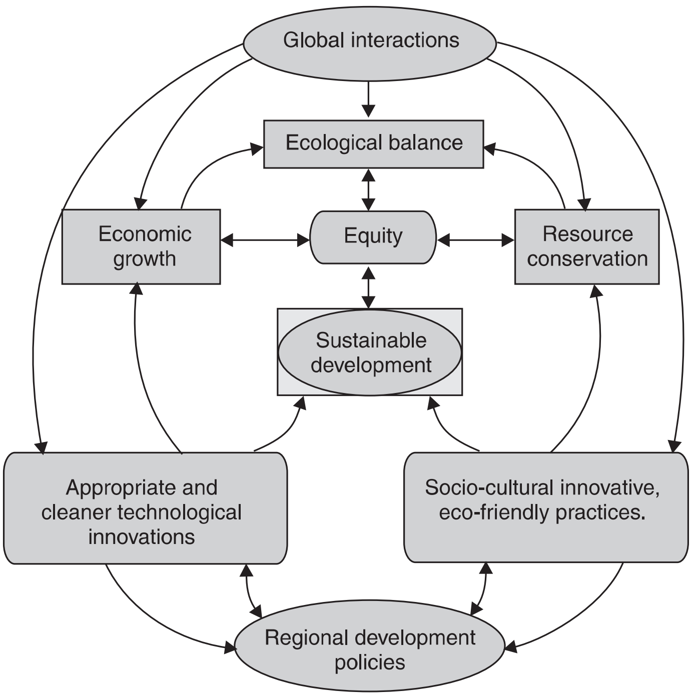

Fig. 3.2. Multidimensional model for sustainable development.

Sustainable development is possible by considering the earth and its resources as common for all. Participatory role of public and different nations for evolving technological innovations and conservationist life style is equally important to achieve economic growth, ecological balance, equity and resource conservation.

I. QUESTIONS

1. What is Environmental Impact Assessment? Why should we have EIA of any developmental project?
2. What are the goals of EIA?
3. Draw a flow chart of EIA methodology.
4. What is Environmental Impact Statement (EIS)? What are the important points considered for preparing the EIS?
5. What is the significance of Environmental audit?
6. Define sustainable development? Why is it necessary?
7. What are inter-generational and intra-generational equity?
8. Define carrying capacity of a system.
9. What is included in the baseline data?

II. OBJECTIVE TYPE QUESTIONS

(A) FILL IN THE BLANKS

1. EIA guidelines were first of all provided by ....
2. .... is an important step in EIA which helps to avoid, reduce or rectify the adverse changes due to the project.
3. The concept of sustainable development was given by ....
4. The Earth Summit was held at ....
5. .... is the capacity to tolerate different stresses.

(B) CHOOSE THE CORRECT ANSWER

1. Gazette notification on EIA is issued by
(*a*) Ministry of Sports
(*b*) Ministry of Human Resource Development
(*c*) Ministry of Finance
(*d*) Ministry of Environment and Forests.

2. EIA helps to
(*a*) achieve sustainable development
(*b*) reap environmental benefits without its degradation

(*c*) maintain balance between population and resources
(*d*) all of these.

3. Study of various atmospheric parameters is called
(*a*) Geology
(*b*) Hydrology
(*c*) Meteorology
(*d*) None of these.

4. UN Conference on Environment and Development (UNCED) is popularly known as
(*a*) Montreal protocol (*b*) Basel conference
(*c*) Earth summit (*d*) None of these.

5. Capacity of a system to sustain a maximum number of organism on a long-term basis is known as
(*a*) Buffering capacity (*b*) Carrying capacity
(*c*) Limited capacity (*d*) None of these.

(C) WRITE TRUE OR FALSE

1. Quality of life on this planet can be maintained by following the principles of sustainable development. (True/False)

2. EIA procedure takes care of projects during their production stage only. (True/False)

3. Environmental Impact Statement includes data on air pollution and water pollution only. (True/False)

4. Reduce, Reuse, Recycle approach will not make much difference in environmental protection. (True/False)

5. Agenda-21 was proposed during the Earth Summit held at Rio de Janeiro. (True/False)